# Mohammad Amin Talaghat, Ph.D.

**Head of Technical Office & Contracts | Contracts & Technical Affairs Manager** |
Ph.D. Construction Management (Amirkabir) | Digital Twin & AI

Ph.D. (Amirkabir University of Technology) and DigiPave researcher at the University of Oulu, Finland, bridging Q1-published research in AI-driven digital twins with a decade of hands-on leadership in construction, technical office and contract management.


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| EOT | 10+ | Q1 | 72 | 3,300 t/d | 1,210 |
|---|---|---|---|---|---|
| DELAY CLAIMS · CONTRACTS | YEARS IN INDUSTRY | ELSEVIER · NATURE · ASCE | CITATIONS · H-INDEX 4 | CEMENT PLANT CLAIM | HOUSING UNITS CONTROLLED |

## PROFESSIONAL PROFILE

Civil engineer and construction management professional who combines **international research in AI-driven digital twins** with **senior industry experience** in project controls, technical office leadership and contract administration. Co-founder and CEO of a construction-contracting company and Head of Technical Office & Contracts at Sahar Sazeh Novin Paydar. Experienced in delay and disruption claims, quantity surveying, interim and final payment statements, price adjustment and new-rate analysis, CPM scheduling and earned value management. Ph.D. research produced an implemented road pavement digital twin for the City of Oulu, Finland, published in *Automation in Construction*, *Scientific Reports*, *ASCE Journal of Computing in Civil Engineering* and the *International Journal of Pavement Engineering*. Builds practical tools across BIM, computer vision, data analytics and agentic-AI automation to deliver faster, verifiable engineering outcomes.

## PROFESSIONAL EXPERIENCE

### Head of Technical Office & Contracts | Contracts & Technical Affairs Manager
**Jun 2025 – Present**

**Sahar Sazeh Novin Paydar** · Shiraz, Iran

- Leads the technical office and contract function: correspondence, progress reporting, estimation, contract records and claim evidence.
- Prepares and reviews quantity take-offs, interim payment statements, variations, new-rate and price analyses, and project control deliverables.
- Prepared the delay and damages claim and the schedule update for the 3,300 t/day Fars Cement Factory (Kharameh).

### Ph.D. Researcher / Research Collaborator — DigiPave
**Jan 2024 – Present**

**University of Oulu** · Finland (hybrid) · Visiting Researcher on site, Oct 2024 – Apr 2025

- Designed and implemented an AI-driven road pavement digital twin and maintenance decision-support system for the City of Oulu.
- Integrated georeferenced inspection data, computer vision, deep learning, BIM/IoT concepts and maintenance optimization into deployable workflows.

### Co-Founder, CEO & Project Manager
**2019 – Present**

**ArkaNoyan Industry Development Co.** · Shiraz, Iran

- Directs construction-contracting and engineering operations: execution, planning, cost control, financing and commercial follow-up.
- Leads multidisciplinary teams across the technical, financial and contractual aspects of building and infrastructure projects.

## CORE EXPERTISE

Project Controls · Contract Management · Delay & Claims · Digital Twin · AI / Deep Learning · Quantity Surveying · Cost Estimation · Payment Statements · Price Adjustment · CPM & EVM · BIM / IoT · Computer Vision · Pavement Management · Asset Management · Structural Design · Risk & BOT Finance · QA/QC · HSE · Tender Documents

## SOFTWARE

### PLANNING & COST

| | |
|---|---|
| MS Project | ●●●●● |
| Primavera P6 | ●●●●○ |
| Taksa (QS & statements) | ●●●●● |
| MS Excel · VBA · Office | ●●●●● |

### BIM · CAD · STRUCTURAL

| | |
|---|---|
| AutoCAD | ●●●●○ |
| Revit (BIM) | ●●●●○ |
| ETABS · SAFE · SAP2000 | ●●●●○ |
| Tekla Structures | ●●●○○ |
| Civil 3D · InfraWorks | ●●●○○ |
| Blender · Photoshop · XD | ●●●○○ |

### DATA · AI · CODE

| | |
|---|---|
| Python (PyTorch, scikit-learn) | ●●●●○ |
| YOLO · OpenCV · Diffusion | ●●●●○ |
| MATLAB · RapidMiner · Weka | ●●●○○ |
| React · PHP/Laravel · JS | ●●●●○ |

## EXPERIENCE (CONTINUED)

### Founder & Director — Web and Software Platforms
**2016 – Present**

**WebAlfa** and **Tarnamagostar Pars Labkhand** · Shiraz, Iran

- Designs and develops web platforms and engineering applications (UI/UX, React, PHP/Laravel, Python), including WebAlfa, BIMGate, Paprika and TwinUp.

### Technical Office & Project Controls Supervisor
**Apr 2025 – Jun 2025**

**Arshsazan Negin** · Shiraz, Iran

- Supervised technical office workflows, quantity surveying, scheduling, documentation and site–design–management coordination.

### Contract Affairs Expert
**Jan 2022 – Jan 2024**

**Iran Gas Engineering & Development Co. (NIGC Eng.)** · Tehran, Iran

- Contract affairs, risk analysis, project finance and BOT studies for oil, gas and infrastructure projects; research on fair risk allocation and contractual structures.

### Technical & Execution Supervisor
**2020 – 2021**

**Peynavardan Shahr Co.** · Shiraz, Iran

- Supervised execution, site coordination, technical documentation and quality follow-up for construction works.

### Technical Office & Project Controls Engineer
**2018 – 2019**

**Debleh Construction Co.** · Tehran, Iran

- Prepared schedules, progress control documents, quantity surveying reports, technical submittals and project coordination documents.

### Executive Engineer · Site Supervisor
**2015 – 2017**

**Peynavardan** · **Borje Korneshe Foolad** · **Tarh-e Mana Shahr**

## EDUCATION

### Ph.D., Civil Engineering — Construction Engineering & Management
**2021 – Jan 2026**

Amirkabir University of Technology (Tehran Polytechnic) · Dept. of Civil & Environmental Engineering

*Dissertation: Development of a digital twin-based intelligent decision support system for asphalt pavement maintenance.* Supervisors: Dr. Amir Golroo, Dr. Vahid Shahhosseini.

### Visiting Researcher (Funded Fellowship) — DigiPave Project
**2024 – 2025**

University of Oulu, Finland · Centre for Wireless Communications · Road pavement digital twins and AI-driven infrastructure monitoring.

### M.Sc., Civil Engineering — Construction Engineering & Management
**2017 – 2020**

K. N. Toosi University of Technology · Faculty of Civil Engineering (Ranked 3rd in Cohort)

*Thesis: A building energy management system integrating BIM and IoT to improve efficiency in public buildings.*

### B.Sc., Civil Engineering
**2011 – 2016**

Shiraz University

## SECTOR EXPERIENCE

| | | |
|---|---|---|
| 01 Industrial plants & cement | 02 Oil, gas & energy infrastructure | 03 Mass housing & mixed-use |
| 04 Roads, pavements & railways | 05 University & public buildings | 06 Smart cities & digital platforms |

## AGENTIC AI

- LLM agents (Claude Code)
- Model Context Protocol
- Agentic CAD/BIM automation
- AutoCAD · Revit · Tekla via MCP
- MSP · Excel automation
- RAG & knowledge bases
- Prompt engineering
- Document AI for contracts

## LANGUAGES

Persian — Native ●●●●●

English — Professional ●●●●○

## HONORS

**National Elites Foundation Award**
Talented students' award, Iran, 2021

**Ranked 20th nationwide**
Iranian Ph.D. Entrance Exam, 2021

**Ranked 3rd**
M.Sc. cohort, Construction Engineering & Management, 2020

**Ph.D. Visiting Scholar**
Funded fellowship, University of Oulu, 2024–2025

## LICENSES

**Licensed Construction Supervisor & Executor (Grade 3)**
Iran Construction Engineering Organization

## TRAINING

- HSE
- ETABS / SAFE design
- Revit / BIM
- AutoCAD
- MATLAB programming
- Python data mining & ML
- MS Project
- Taksa QS & statements

## SELECTED PROJECTS

**Road Pavement Digital Twin — DigiPave**
Decision-support system for pavement maintenance, City of Oulu, Finland (Business Finland).

**Fars Cement Factory · 3,300 t/day**
Delay and damages claim and project schedule, Kharameh, Iran.

**Q Mixed-Use Complex · 170,000 m²**
Delay claim documentation, Khorramabad, Iran.

**Mehr Housing · 1,210 units · 120,000 m²**
Project planning and controls, Pardis City.

**60-ha Site Works · Mehr Housing**
Quantity surveying and cost estimation, Pardis City.

**BIM/IoT Energy Management**
Smart-building energy system for Roger Williams College, Florida, USA.

**Railway Condition Assessment**
YOLO-based vision and georeferenced maintenance intelligence, Zanjan–Tehran line.

**Shiraz University ECE Faculty · 6,000 m²**
Construction of the Electrical & Computer Engineering Faculty building.

**6-Storey Residential · 2,000 m²**
Architectural drawings and structural design, Shiraz.

**Digital Platforms**
WebAlfa, BIMGate, Paprika and TwinUp: design and development.

## CODES & STANDARDS

General Conditions of Contract
Circulars 5090 · 173073 · 4951
Iranian National Building Code
Standard 2800 (seismic)
ASTM D6433 (PCI)
AASHTO 1993
PMBOK
DCMA 14-point

## RESEARCH FOCUS

Infrastructure digital twins
Pavement distress segmentation
PCI estimation
Generative & self-supervised AI
Maintenance optimization
Explainable deterioration models
BIM/IoT energy
Contract risk allocation

## CONSULTING & SERVICES

**Claims & Contracts**
Delay and disruption claims, EOT, contract review, variations, dispute support.

**Technical Office**
Quantity take-off, estimates, interim/final statements, price adjustment, new rates.

**Project Controls**
WBS, CPM schedules in MSP/P6, EVM, S-curves, progress and PMO reporting.

**BIM & CAD**
Revit modelling, AutoCAD/Tekla drawings, MTO, IFC-based coordination.

**AI & Digital Twin**
Computer vision, predictive models, dashboards and decision support for assets.

**Research Support**
Q1 manuscripts, data analysis, code implementation, revision and peer response.

## DIGITAL & AI CAPABILITIES

### Agentic Engineering Automation

- LLM agents (Claude Code) orchestrating AutoCAD, Revit, Tekla, MS Project and Excel through Model Context Protocol (MCP) connectors
- Automated take-off, scheduling and reporting pipelines with human review gates
- Document AI for contracts, correspondence and claim evidence

### Computer Vision & Digital Twins

- Distress detection and segmentation (YOLO, diffusion-based augmentation, self-supervised learning)
- Automated crack quantification and PCI estimation (ASTM D6433)
- Georeferenced asset data fused into digital-twin decision support

### Data, Platforms & Decision Support

- Predictive deterioration and maintenance-optimization models (MLP–LSTM, explainable AI)
- Web platforms and dashboards: WebAlfa, BIMGate, Paprika, TwinUp
- Python, React and Laravel applications for engineering teams

## PUBLICATIONS

1. Talaghat, M. A., Golroo, A., Shahhosseini, V., & Rasti, M. (2026). **An implemented AI-driven digital twin architecture for proactive pavement maintenance in Finland.** *Scientific Reports*, 16, 24074. doi:10.1038/s41598-026-46719-z Q1
2. Talaghat, M. A., Golroo, A., Shahhosseini, V., Rasti, M., Daneshvar, A., & Moghadas Nejad, F. (2026). **Hybrid MLP–LSTM deep predictive modeling of pavement distress severity and density and condition indices.** *International Journal of Pavement Engineering*, 27(1), 2624565. doi:10.1080/10298436.2026.2624565 Q1
3. Talaghat, M. A., Golroo, A., Rasti, M., & Shahhosseini, V. (2026). **Road Pavement Digital Twin: A Blueprint with a Case Study.** *Journal of Computing in Civil Engineering*, 40(3), 05026001. doi:10.1061/JCCEE5.CPENG-7187 Q1
4. Talaghat, M. A., Golroo, A., Shahhosseini, V., & Rasti, M. (2025). **Enhancing pavement distress segmentation model with diffusion-based generative AI: a digital twin perspective.** *International Journal of Pavement Engineering*, 26(1), 2540076. doi:10.1080/10298436.2025.2540076 Q1
5. Talaghat, M. A., Golroo, A., Kharbouch, A., Rasti, M., Heikkilä, R., & Jurva, R. (2024). **Digital twin technology for road pavement.** *Automation in Construction*, 168, 105826. doi:10.1016/j.autcon.2024.105826 Q1
6. Talaghat, M. A., Golroo, A., Shahhosseini, V., & Rasti, M. (2026). **Fully Automated Pavement Crack Quantification and PCI Estimation via a Digital Twin-Ready Vision-Based Deep Learning Pipeline.** *Results in Engineering*, 32, 111659. doi:10.1016/j.rineng.2026.111659 Q1
7. Talaghat, M. A., Golroo, A., Shahhosseini, V., Rasti, M., & Daneshvar, A. (2026). **A data-driven optimization framework for network-level pavement maintenance planning.** *International Journal of Pavement Engineering*. Under Review
8. Sedighian-Fard, M., Golroo, A., Javanmardi, M., Alahi, A., Dehghan Tezerjani, H., Talaghat, M. A., Shahhosseini, V., & Rasti, M. (2026). **Explainable AI–enhanced pavement deterioration modeling using hybrid classification–regression deep learning architecture.** *Scientific Reports*. doi:10.1038/s41598-026-67842-x Q1
9. Nouri, S., Rezaeemanesh, M., Talaghat, M. A., Rasti, M., & Golroo, A. (2026). **Self-supervised crack segmentation with data-driven severity detection.** *IEEE Transactions on Intelligent Transportation Systems*. Under Review
10. Sedighian-Fard, M., Talaghat, M. A., et al. (2026). **Deep vision-based railway track superstructure condition assessment using YOLOv26 and georeferenced maintenance intelligence: Field validation on the Zanjan-Tehran railway line.** *Scientific Reports*. Under Review
11. Sedighian-Fard, M., Golroo, A., Javanmardi, M., Alahi, A., Talaghat, M. A., Shahhoseini, V., Amiri, S., & Moghadas Nejad, F. (2026). **A complete pipeline for automated pavement distress detection, segmentation, and quantification using a new surface condition index.** *Case Studies in Construction Materials*. Under Review
12. Sedighian-Fard, M., Javanmardi, M., Alahi, A., Talaghat, M. A., & Shahhosseini, V. (2026). **Vehicle-based pavement condition monitoring: A review of evidence, technology readiness, and management integration.** *Journal of Traffic and Transportation Engineering*. Under Review
13. Alobaidi, M., Golroo, A., Sedighian-Fard, M., & Talaghat, M. A. (2026). **Synchronized 2D–3D edge sensing for automated ASTM D6433 pavement condition assessment: A Baghdad case study.** *Measurement*. Under Review
14. Talaghat, M. A., Sedighian-Fard, M., Golroo, A., & Rasti, M. (2025). **Leveraging digital twin technology for data-driven pavement maintenance.** *AM3P 2025 Conference*, Vol. 4, pp. 604–608. TU Wien. doi:10.34726/10886 Conference
15. Talaghat, M. A., Golroo, A., & Rasti, M. (2025). **Improving Pavement Distress segmentation with Diffusion-Based Generative AI.** *AM3P 2025 Conference*, Vol. 4, pp. 579–582. TU Wien. doi:10.34726/10798 Conference
16. Sedighian-Fard, M., Fathali, M., Yazdani Tabai, S. H., Homavandi, S., Akbari, M., & Talaghat, M. A. (2026). **Interpretable machine-vision indices for monitoring sand fouling, ballast whitening and vegetation in climatically diverse Iranian railways.** *International Journal of Railway Research*. Under Review
17. Saghatforoush, E., Abbasianjahromi, H., Talaghat, M. A., & Kahvandi, Z. (2022). **Development of a Management System to Improve the Energy Efficiency of Public Buildings by Integrating IoT and BIM.** *Lean Construction Journal*, 82–104. Peer-Reviewed
18. Talaghat, M. A., & Sedighian-Fard, M. (2026). **A self-calibrating probabilistic digital twin of road pavement networks with decision-centric tasking of autonomous vehicle fleets.** Under Review
19. Talaghat, M. A., & Sedighian-Fard, M. (2026). **A self-updating, two-way building digital twin for grid-interactive operation: Forecasting equipment faults and jointly optimising energy, peak demand, carbon and maintenance.** *Sustainable Energy, Grids and Networks*. Under Review
20. Talaghat, M. A., & Sedighian-Fard, M. (2026). **ARGONet-4D: A maturity-graded digital twin of urban road surfaces that links fleet-sourced 4D reconstruction, foundation-model distress objects, IFC 4.3 semantics, IoT-informed forecasting and prescriptive maintenance.** *Innovative Infrastructure Solutions*. Under Review

## REFERENCES

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